Scalability & Future Plan

Date	14 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI Intelligent Crop Recommendation System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Prediction is based only on the trained dataset and does not use live weather information.	Recommendations may not reflect real-time environmental conditions.	High
2	Supports crop recommendation only and does not include fertilizer or pest management suggestions.	Limited decision support for farmers.	Medium
3	The application is intended for moderate usage and has not been optimized for large-scale concurrent users.	Performance may decrease under heavy traffic.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports individual and small-scale users.	Deploy on cloud infrastructure with load balancing and auto-scaling support.
2	Data Storage	Local dataset and storage.	Integrate cloud database (MongoDB Atlas/PostgreSQL) for scalable storage.
3	Performance	Optimized for prototype demonstration.	Add caching, API optimization, and model inference improvements.

4	Security	Basic application security.	Implement user authentication, HTTPS, role-based access control, and API security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Weather API Integration for real-time recommendations	1 month	More accurate crop prediction
Phase 2	Fertilizer Recommendation System	2monnths	More accurate crop prediction
Phase 3	Crop Disease Detection using Deep Learning	3months	Improved accessibility and scalability